

DATE:

EXPT NO: 02

EDA - DATA CLEANING

AIM:

To perform Exploratory Data Analysis (EDA) and data cleaning on the Earthquake dataset to identify patterns, handle missing values, remove duplicates, and prepare the data for Machine Learning.

PROCEDURE:

1. Import the required Python libraries such as Pandas, NumPy, Matplotlib, and Seaborn.
2. Load the Earthquake dataset using Pandas.
3. Display the first few records and check the dataset shape.
4. Examine the data types and dataset information.
5. Check for missing or null values.
6. Handle missing values using suitable methods.
7. Identify and remove duplicate records.
8. Check for incorrect or inconsistent data.
9. Perform statistical analysis using describe().
10. Visualize important features using graphs and charts.
11. Analyze correlations between numerical features using a heatmap.
12. Save the cleaned dataset for further analysis and Machine Learning.

CODE:

```
import pandas as pd

# Import Excel file
df = pd.read_excel(r"C:\Users\babyn\Downloads\archive\earthquake_data.xlsx")

# Display first 5 records
print("First 5 Records")
print(df.head())

# Dataset Information
print("\nDataset Information")
df.info()

# Statistical Summary
print("\nStatistical Summary")
```

```
print(df.describe(include='all'))
```

OUTPUT:

First 5 Records

								title	magnitude	date_time	cdi	mmi	alert
tsunami	sig	net	nst	dmin	gap	magType	depth	latitude	longitude				
location	continent			country									
0	M	7.0	-	18 km SW of Malango, Solomon Islands			7.0	22-11-2022 02:03	8	7	green		
1	768	us	117	0.509	17.0	mww	14.000	-9.7963	159.596	Malango, Solomon Islands			
	Oceania			Solomon Islands									
1	M	6.9	-	204 km SW of Bengkulu, Indonesia			6.9	18-11-2022 13:37	4	4	green		
0	735	us	99	2.229	34.0	mww	25.000	-4.9559	100.738	Bengkulu, Indonesia			
	NaN			NaN									
2						M 7.0 -	7.0	12-11-2022 07:09	3	3	green		
1	755	us	147	3.125	18.0	mww	579.000	-20.0508	-178.346				NaN
	Oceania			Fiji									
3	M	7.3	-	205 km ESE of Neiafu, Tonga			7.3	11-11-2022 10:48	5	5	green		
1	833	us	149	1.865	21.0	mww	37.000	-19.2918	-172.129	Neiafu, Tonga			
	NaN			NaN									
4						M 6.6 -	6.6	09-11-2022 10:14	0	2	green		
1	670	us	131	4.998	27.0	mww	624.464	-25.5948	178.278				NaN
	NaN			NaN									

Dataset Information

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 782 entries, 0 to 781
Data columns (total 19 columns):
#   Column      Non-Null Count  Dtype
---  -
0   title       782 non-null    object
1   magnitude   782 non-null    float64
2   date_time   782 non-null    object
3   cdi         782 non-null    int64
4   mmi         782 non-null    int64
5   alert       415 non-null    object
6   tsunami     782 non-null    int64
7   sig         782 non-null    int64
8   net         782 non-null    object
9   nst         782 non-null    int64
10  dmin        782 non-null    float64
11  gap         782 non-null    float64
12  magType     782 non-null    object
13  depth       782 non-null    float64
14  latitude    782 non-null    float64
15  longitude   782 non-null    float64
16  location    777 non-null    object
17  continent   206 non-null    object
18  country     484 non-null    object
dtypes: float64(6), int64(5), object(8)
memory usage: 116.2+ KB
```

Statistical Summary

	magnitude	cdi	mmi	tsunami	sig	nst	dmin
gap	depth	latitude	longitude				

```

count    782.000000    782.000000    782.000000    782.000000    782.000000    782.000000    782.000000
782.000000    782.000000    782.000000    782.000000
mean      6.940153     4.333760     5.964194     0.388747    776.008951    275.396419     2.231937
25.807864    75.883168     2.386000     39.734707
std       0.445213     2.658939     1.983592     0.487778    185.083392    218.192074     2.839655
17.652131    137.279603     30.344421    108.016952
min       6.500000     0.000000     1.000000     0.000000    650.000000     0.000000     0.005000
9.000000     9.000000    -61.848000   -179.962000
50%       6.800000     4.000000     6.000000     0.000000    708.500000    213.000000     1.300000
21.000000    26.000000    -3.580000    103.982000
75%       7.100000     7.000000     7.000000     1.000000    819.750000    385.000000     2.880000
29.000000    55.000000    26.126500    139.710250
max       9.100000     9.000000     9.000000     1.000000   2910.000000    934.000000    17.654000
239.000000   670.810000    71.631200    179.662000
[11 rows x 19 columns]

```

```

import pandas as pd

df = pd.read_csv(r"C:\Users\babyn\Downloads\archive\earthquake_data.csv")
print(df.head())

```

OUTPUT:

```

              title magnitude      date_time  cdi  mmi  alert
tsunami sig net nst  dmin  gap magType  depth latitude longitude
location continent          country
0  M 7.0 - 18 km SW of Malango, Solomon Islands      7.0  22-11-2022 02:03     8     7  green
1  768  us  117  0.509  17.0      mww  14.000  -9.7963   159.596  Malango, Solomon Islands
Oceania Solomon Islands
1      M 6.9 - 204 km SW of Bengkulu, Indonesia      6.9  18-11-2022 13:37     4     4  green
0  735  us   99  2.229  34.0      mww  25.000  -4.9559   100.738    Bengkulu, Indonesia
NaN              NaN
2              M 7.0 -      7.0  12-11-2022 07:09     3     3  green
1  755  us  147  3.125  18.0      mww  579.000  -20.0508  -178.346              NaN
Oceania              Fiji
3      M 7.3 - 205 km ESE of Neiafu, Tonga      7.3  11-11-2022 10:48     5     5  green
1  833  us  149  1.865  21.0      mww  37.000  -19.2918  -172.129    Neiafu, Tonga
NaN              NaN
4              M 6.6 -      6.6  09-11-2022 10:14     0     2  green
1  670  us  131  4.998  27.0      mww  624.464  -25.5948   178.278              NaN
NaN              NaN

[5 rows x 19 columns]

```

```

import pandas as pd
import sqlite3

# Read Excel file
df = pd.read_excel(r"C:\Users\babyn\Downloads\archive\earthquake_data.xlsx")

```

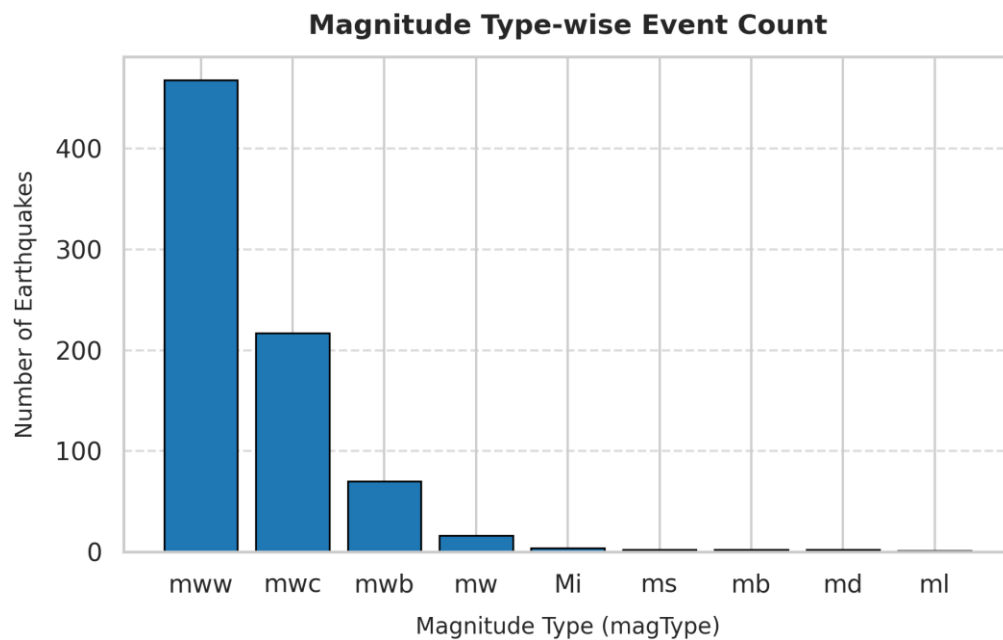
```
# Connect to SQLite database
conn = sqlite3.connect("earthquake_data.db")

# Export data to SQLite table
df.to_sql("earthquake_data", conn, if_exists="replace", index=False)
print("Data exported to SQLite successfully!")
conn.close()
```

```
import matplotlib.pyplot as plt

mag_type = df["magType"].value_counts()
plt.bar(mag_type.index, mag_type.values)
plt.title("Magnitude Type-wise Event Count")
plt.xlabel("Magnitude Type (magType)")
plt.ylabel("Number of Earthquakes")
plt.show()
```

OUTPUT:



RESULT:

The Earthquake dataset was successfully cleaned and analyzed using EDA. Missing values and duplicate records were handled, important patterns were identified through visualizations, and the cleaned dataset was prepared for further Machine Learning analysis.